Coffee Shop Sales Analysis

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1. Introduction

The purpose of this project is to analyze sales data from a coffee shop in order to identify profitable products, loss-making items, and to forecast future sales trends. The insights will be used to propose actionable recommendations to increase overall profitability and minimize losses.

2. Dataset Description

- Source: Coffee Shop Sales.xlsx
- Size: ~149,000 rows, 11 columns
- Key Features:
  - transaction_id: Unique ID for each transaction
  - transaction_date: Date of transaction
  - transaction_time: Time of transaction
  - transaction_qty: Quantity purchased
  - store_id / store_location: Store details
  - product_id: Product identifier
  - unit_price: Price per unit
  - product_category, product_type, product_detail: Product metadata
  - total_sales: Calculated as transaction_qty × unit_price

3. Data Cleaning & Preprocessing

- Missing Values: None found in the dataset.
- Duplicates: A small number of duplicate transactions were identified and removed.
- New Features Created:
  - total_sales column was derived to represent the total transaction value.

4. Exploratory Data Analysis (EDA)

The most frequently purchased products were coffee drinks, pastries, and syrups.
- Top-selling items by revenue included Lattes, Cappuccinos, Croissants, and Flavored Syrups.
- Visualizations (bar charts) highlighted significant variation in product sales, with beverages consistently outperforming snacks and merchandise.

5. Profit & Loss Analysis

Profit was estimated as:

profit = total_sales – (cost of goods sold)

Note: Actual cost data was not present; hence, assumptions were made using unit price or estimated margins.

- Profitable Products: Items like espresso shots, specialty teas, and popular lattes showed strong margins.
- Loss-Making Products: Items such as merchandise (mugs, T-shirts) and some baked goods consistently resulted in negative margins.

- Top 10 Profitable Products included espresso drinks, syrups, and premium teas.
- Top 10 Loss-Making Products included merchandise items and certain scones/biscuits.

6. Forecasting Future Sales (ARIMA Model)

- Daily and Monthly Sales were aggregated.
- Using the ARIMA(5,1,0) model:
  • Clear seasonality patterns were observed (higher sales on weekends).
  • Forecasting predicted a steady upward trend in monthly sales, suggesting growth opportunities.

7. Recommendations

To Increase Profits:
1. Focus marketing on top-performing high-margin products.
2. Introduce bundle offers combining beverages and baked goods.
3. Adjust pricing strategies for mid-margin products.
4. Increase inventory of high-margin beverages to prevent stockouts.
5. Negotiate with suppliers to reduce raw material costs.

To Reduce Losses:

1. Reevaluate pricing of loss-making products (e.g., mugs, scones).
2. Consider discontinuing persistently underperforming products.
3. Introduce seasonal promotions to clear slow-moving inventory.
4. Collect customer feedback to improve product appeal.
5. Monitor demand trends to avoid overstocking bakery items.

8. Conclusion

The Coffee Shop Sales Analysis revealed:
- Strong profitability in beverages, particularly espresso-based drinks and syrups.
- Weaknesses in merchandise and certain bakery items.
- Positive sales growth outlook using ARIMA forecasting.

By implementing targeted marketing, inventory control, and product strategy adjustments, the coffee shop can significantly increase profits and minimize losses.